



Stock pledge, risk of losing control and corporate innovation

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ABSTRACT

This paper investigates the effects of stock pledge by controlling shareholder on corporate's future innovation productivity and the mechanism through which stock pledge affects innovation. We find that both the existence of stock pledge by controlling shareholder and the percentage of shares pledged by controlling shareholder are significantly negatively related to firms' future innovation outputs and quality, and these baseline results are robust to a variety of tests on sample selections, model specifications, and variable definitions. We further adopt several methodologies to address endogeneity concerns and establish a causal relationship between stock pledge by controlling shareholder and innovation. We then provide evidence to show that the impediment effect of stock pledge by controlling shareholder on innovation is possibly due to controlling shareholder's fear of losing corporate control in case of innovation failure. Finally, we find that although stock pledge is a possible channel to relieve a firm's financial constraint, it does not encourage the firm to invest more in innovation.

1. Introduction

Over the past decade, pledging shares by the controlling shareholder in exchange for loans becomes an increasingly prevalent financing channel for listed firms in China. According to a report issued by China Securities Depository & Clearing Corp (CSDC), till the end of 2017, the controlling shareholders of 1951 A-share listed firms had pledged their shares, which accounted for about 10% of total shares outstanding in the Chinese market with a market value exceeding 5 trillion RMB. The rapid growth of the stock pledge has attracted extensive attention from investors, media, and regulators. Although a few studies have attempted to investigate the risk involved in stock pledge and the effects of stock pledge on near-term firm value and shareholders' wealth (Chan et al., 2018; Dou et al., 2019), the economic consequence of stock pledge on the firm's long-term growth is still under-explored. The objective of this paper is to provide the first empirical study that investigates how stock pledge by controlling shareholder affects corporate innovation, which is a key driver of the long-term competitiveness of a firm and even a country.

Despite the fact that there has been a growing literature relating various market and firm characteristics to innovation (Hirshleifer et al., 2012; Aghion et al., 2013; He and Tian, 2013; Tian and Wang, 2014; Fang et al., 2014; Cornaggia et al., 2015; Jiang and Yuan, 2018), little is known about the effect of the stock pledge. Understanding the role of the stock pledge in motivating innovation is important because whether corporate insiders should be allowed to pledge their shares has caused heated debate both in China and other countries.¹ The reason we focus

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¹ For example, a survey conducted by Institutional Shareholder Service (ISS) finds that 49% of institutional shareholders in the U.S. thought that pledging of shares by corporate insiders is problematic and states that "Pledging of company stock in any amount is not a responsible use of equity" (Institutional Shareholder Service, 2012). However, the advocates of stock pledge argue that prohibiting stock pledge will cause corporate insiders to reduce their ownership, which may induce more agency problems.

on stock pledge by controlling shareholder rather than other blockholders in this study is that controlling shareholders are the ultimate decision-makers of the firms in China. Therefore, their risk-taking willingness directly determines the size of investments in innovation projects. Given that controlling shareholders are common around the world (La Porta et al., 1999; Claessens et al., 2000; Faccio and Lang, 2002; Anderson et al., 2009), the implications of our paper are also applicable to other countries.

The stock pledge may have two opposite effects on innovation incentives of the controlling shareholder. On the one hand, stock pledge not only allows the controlling shareholder to cash out up to 100% of their ownership, but also allows them to benefit from any stock price appreciation in the future. Dou et al. (2019) describe the payoff structure of stock pledge as a “call option like” because once the value of the shares pledged falls below the value of the loan, the controlling shareholder can default and walk away from his loan obligations. Therefore, the controlling shareholder's downside risk is effectively hedged, but the upside potential is not limited, which provides him with an incentive to take more risk and pursue more innovative projects.

On the other hand, Holmstrom (1989) points out that innovation is a highly uncertain process with a high probability of failure. The failure of innovation projects may cause stock price slump and trigger the margin call of a stock pledge. A default in the pledged loan will lead to a forced share sale, which brings a significant risk of losing corporate control to the controlling shareholder.² The above risk-taking incentive argument assumes that controlling shareholders do not put the safety of their controlling positions in the first priority, but it may not be the case. Barclay and Holderness (1989) find that blockholders in public firms have significant private benefits of control, and Dyck and Zingales (2004) further show that the private benefits of control are more significant with less developed capital markets and more concentrated ownership. To keep their valuable private benefits of control, the controlling shareholders may have incentive to maintain a stable stock price during stock pledge periods and have low tolerance to any event that may destroy firms' short-term performance and stock prices, so as to lower the probability of losing corporate control in case of forced share sales triggered by the margin call. Following this argument, a controlling shareholder may have an incentive to reduce the firm's risk exposure by suspending innovation projects if he has shares under pledge, and this incentive should be more pronounced if more shares are pledged by the controlling shareholder.

Motivated by these two arguments, this paper examines the effects of stock pledge by controlling shareholders on corporate innovation for all A-share listed firms in China from 2003 to 2013. Our baseline results show that both the existence of controlling shareholder's stock pledge and the percentage of shares pledged by the controlling shareholder are significantly negatively related to the firm's future innovation outputs and quality. The results are robust to alternative sample selections, model specifications, and variable definitions.

However, our baseline analyses may subject to endogeneity concerns. First, it is possible that some unobserved variables omitted from our empirical model are correlated with both stock pledge and innovation, rendering our findings spurious (i.e., the omitted variables concern). Second, it is possible that our multiple regression model suffers from functional form misspecification problem which causes our findings to be biased (i.e., the functional form misspecification concern).³ Last, it is also plausible that the controlling shareholders of low innovation potential firms are more likely to pledge their shares (i.e., reverse causality concern). In order to establish causality between the stock pledge and innovation, we adopt three strategies to address endogeneity concerns.

Our first strategy is to control for firm fixed effects throughout our regression analyses to mitigate the concern about the unobserved time-invariant firm characteristics that may affect both stock pledge and innovation. We find that the negative relationship between the stock pledge and innovation is still robust after controlling for firm fixed effects. Our second strategy is to perform nearest-neighbor propensity score matching to mitigate the concern about the functional form misspecification problem. More specifically, we perform a 1-to-1 match of firms whose controlling shareholders have stocks under pledge (treatment group) with firms whose controlling shareholders do not pledge stocks (control group) each year based on their propensity scores. We find that the average number of future patents generated by the treatment group is significantly lower than that generated by the control group. Moreover, the results of multivariable regressions using the matched sample are consistent with our baseline results. Our third strategy is to perform a two-stage-least-squares (2SLS) analysis to mitigate the concerns about reverse causality and other potential omitted variables. We construct an instrumental variable as the “average percentage of shares pledged by the controlling shareholders of non-innovative firms (firms never generate any patent) that are located in the same province and belong to the same industry”. The 2SLS regression results confirm the negative relationship between the stock pledge and innovation. Overall, our endogeneity tests suggest that controlling shareholder's stock pledge impedes the firm's future innovation.

However, the negative causal relation we observed between controlling shareholder's stock pledge and innovation only reflects an overall effect, but the mechanism through which stock pledge by controlling shareholder impedes innovation is still unclear. We further examine whether the impediment effect of stock pledge on corporate innovation is possibly due to the controlling shareholder's fear of losing corporate control.

We propose two different scenarios that may exacerbate controlling shareholder's concerns about losing corporate control and hypothesize that if the impediment effect of stock pledge on innovation is caused by the controlling shareholder's fear of losing corporate control, it should be more pronounced in these scenarios. The first scenario is that the controlling shareholder does not hold a wide lead in ownership over the second-largest shareholder, so that the second-largest shareholder potentially challenges his controlling position. Therefore, he may tend to make more conservative corporate decisions that substantially reduce the firm's risk

² One most recent example is about Mr. Jia Yueting, the founder, CEO, Chairman and the largest shareholder of LeTV, which was a star company in China Growth Enterprise Market. From 2015 to 2017, Mr. Jia had pledged 99.53% of his LeTV shares to several financial institutions in exchange of loans for the innovative LeEco and electric-car projects. Unfortunately, both projects failed and the stock price of LeTV slumped by 69% in 11 trading days. On March 1st, 2018, LeTV announced that all of the shares pledged by Mr. Jia are frozen and will be sold through a judicial sale, and Mr. Jia will lose his control of the company.

³ More details about functional form misspecification issue can be found in Rosenbaum and Rubin (1983) and Shipman et al. (2016).

exposure to keep his pledged shares safe so as to secure his controlling position. The second scenario is that the controlling shareholder is under increased stock price crash pressure. If the stock price crash risk of a firm increases significantly when the controlling shareholder pledges his shares, the controlling shareholder should be more conservative in investing in innovation projects because innovation activities have a high failure rate and tend to generate more bad news, which may accelerate the stock price crash (He and Wong, 2004; Jia, 2018). A stock price slump during the stock pledge period may trigger the margin call of pledged loans and threaten the controlling rights of the controlling shareholder. Consistent with our conjecture, we find that the impediment effect of stock pledge by controlling shareholder on innovation becomes more pronounced in both higher controlling power competition and large increase in stock price crash risk subsamples. These findings suggest that the stock pledge impedes corporate innovation because of controlling shareholder's fear of losing corporate control.

Moreover, it is worth to note that although on average, stock pledge by controlling shareholder impedes corporate innovation, we cannot exclude the possibility that stock pledge may foster innovation under certain circumstances. Innovation requires substantial investments in both tangible and non-tangible assets so that firms are less likely to undertake large innovation projects by using only internal financing (Hall, 2002; Acharya and Xu, 2017). Therefore, financial constraints impede investments in innovation projects and the growth of firms (Hubbard, 1998; Stein, 2003; Li, 2011; Nanda and Nicholas, 2014). Compared to traditional external financing, stock pledge offers firms an easier, faster, and less costly financing channel which may help financially constrained firms to access to more external financing and pursue more innovative projects. If this financial constraint relaxation hypothesis is correct, we are supposed to observe a positive or at least, a less negative relationship between stock pledge by controlling shareholder and innovation for financially constrained firms.

Following Hadlock and Pierce (2010), we construct a financial constraints index to measure the level of financial constraints for each firm-year. We find that stock pledge by controlling shareholder always impedes innovation in both financially constrained and non-financially constrained subsamples, and the impediment effect is even stronger for financially constrained firms. These findings contradict the prediction of the financial constraint relaxation hypothesis.

One may argue that the stock pledge does not improve innovation for financially constrained firms because a large portion of controlling shareholders pledge their shares for personal loans rather than for corporate loans. If the loans obtained through stock pledge are not invested in the firm, it will not help to relieve the firm's financially constrained situation.⁴ Unfortunately, listed firms in China are not required to disclose the purpose of stock pledges by blockholders so that we are not able to directly investigate this issue. However, since the controlling shareholders of state-owned-enterprise (SOEs) are the governments, it is highly unlikely that the loans obtained through pledging state-owned shares flow into any personal pocket as embezzlement of state funds is a serious crime.⁵ We find that stock pledge also significantly impedes the innovation of more financially constrained SOEs, and this impediment effect is not weaker than what we observe for less financially constrained SOEs. Therefore, we can conclude that although stock pledge is a possible channel to relieve the firm's financial constraints, it does not encourage the firm to invest more in innovation.

The rest of the paper is organized as follows. Section 2 introduces the institutional background of the controlling shareholders and the stock pledge in China, and discusses the related literature and contributions of this study. Section 3 describes the sample selection and presents summary statistics. Section 4 presents the baseline results, robustness tests, and explores the possible underlying mechanism through which controlling shareholder's stock pledge impedes innovation. Section 5 concludes the paper.

2. Institutional background, related literature and contributions

2.1. Institutional background of controlling shareholders and stock pledges in China

Because of the concentrated ownership structure in China, most of the Chinese listed firms have controlling shareholders. According to Article 216 (II) of the Company Law of China, it is not necessary for a shareholder to hold more than 50% ownership in a firm to become the controlling shareholder. One may control a firm as long as his voting rights are sufficient to have a major impact on the resolutions of the board of directors or general meeting. The China Securities Regulatory Commission (CSRC), Shanghai Stock Exchange and Shenzhen Stock Exchange further interpret the meaning of "control" as "to be in a position to decide an enterprise's financial and operation policies and thereby obtain interest from the enterprise's business operation". More specifically, a controlling shareholder can be an entity either holding the largest number of shares of a firm, or in a position to either directly or indirectly exercise more voting rights than the largest shareholder.⁶ But usually in most papers studying Chinese firms, scholars just consider the largest shareholder as the controlling shareholder (for example, Jiang et al., 2010; Tan et al., 2015), because it is rare to see that the largest shareholder gives up his voting rights, or other blockholders accumulate their voting rights together to vote against the largest shareholder in China.

In Panel B of Table 1, we find that the average ownership of the controlling shareholders in our sample is more than 35%, while the average ownership of other blockholders is only around 20%. Therefore, under the institutional background of China, the controlling shareholders tend to be entrenched and powerful. They normally exercise major control over corporate decisions and directly engage in the daily managerial process, and their positions are not easily challenged by either insiders or outsiders.

The block shareholders of listed firms are allowed to pledge their shares to financial institutions in exchange for loans in China

⁴ We thank the referee for pointing out this possibility.

⁵ A regulation issued by the Ministry of Finance of China in 2002 requires that state-owned shares of listed SOEs can only be pledged for loans to listed firms themselves or to their subsidiaries. This regulation is available at http://www.gov.cn/gongbao/content/2002/content_61623.htm.

⁶ An English version of Rules governing the Listing of Shares on Shanghai Stock Exchange can be visited at <http://english.sse.com.cn/laws/framework/c/4801956.docx> and the interpretation of controlling shareholder can be found in page 105.

Table 1

Sample distributions across the years.

This table presents the sample distributions across the years from 2003 to 2014. Panel A reports the distribution of firms with stock pledge. Panel B reports the distribution of the percentage of shares held and pledged by blockholders. The controlling shareholder is the largest shareholder of a firm, and the non-controlling shareholders are the other nine shareholders in the Top 10 Shareholders list of a firm. Panel C reports the distribution of average patents applications and granted per firm. Since we want to examine the impact of stock pledge on the firm's future innovation productivity, our stock pledge data is from 2003 to 2013 and our patent data is from 2004 to 2014, respectively.

Panel A: distribution of firms with stock pledge.					
	No. of firms	No. of firms with stock pledge by controlling shareholders		No. of firms with stock pledge by non-controlling shareholders	
2003	1238	266		246	
2004	1322	327		277	
2005	1331	354		332	
2006	1384	371		317	
2007	1498	377		279	
2008	1566	427		303	
2009	1661	461		308	
2010	1999	528		407	
2011	2275	662		588	
2012	2424	761		611	
2013	2424	907		766	

Panel B: distribution of the percentage of shares held and pledged by shareholders.					
	Percentage of total shares under pledge	Percentage of shares held by controlling shareholders	Percentage of shares pledged by controlling shareholders	Percentage of shares held by non-controlling shareholders	Percentage of shares pledged by non-controlling shareholders
2003	0.121	0.427	0.136	0.186	0.071
2004	0.137	0.420	0.160	0.199	0.076
2005	0.148	0.404	0.171	0.202	0.091
2006	0.151	0.360	0.186	0.201	0.078
2007	0.128	0.358	0.158	0.197	0.060
2008	0.138	0.363	0.169	0.192	0.058
2009	0.140	0.362	0.173	0.191	0.052
2010	0.127	0.362	0.158	0.214	0.053
2011	0.143	0.362	0.174	0.224	0.068
2012	0.160	0.364	0.199	0.224	0.076
2013	0.185	0.359	0.232	0.216	0.092

Panel C: distribution of average patents applications and grants per firm.				
	Number of total patents application	Number of total patents granted	Number of invention patents application	Number of invention patents granted
2004	4.160	3.660	2.386	1.886
2005	6.047	5.191	3.380	2.525
2006	8.157	6.560	4.968	3.372
2007	11.940	9.712	7.674	5.446
2008	14.916	12.006	8.760	5.850
2009	20.591	16.508	11.688	7.605
2010	26.951	21.990	14.014	9.053
2011	34.500	27.873	17.013	10.387
2012	39.717	32.761	19.024	12.068
2013	41.780	33.382	20.922	12.525
2014	50.673	35.039	26.207	10.575

since 2000. The stock pledge market is growing very fast in China in the past decade, because compared with traditional collateral loans, stock-pledging loans provide specific benefits to both the shareholders and financial institutions. From the shareholders' perspectives, stock-pledging loans are faster, less costly, and easier to obtain, especially after CSRC permitted brokerage companies to participate in stock-pledging loan business in 2013.⁷ From the financial institutions' perspectives, compared with other collaterals, stocks of listed firms are much more liquid, and their market values are observable. By adopting several credit enhancement practices, such as overcollateralization, marking-to-market, and maintenance requirement, financial institutions can minimize their risk exposure in case that the pledgers cannot repay the loans at maturity or the value of pledged shares decreases significantly before the

⁷ Prior to 2013, only banks and trust companies are allowed to do stock-pledging loan business.

loan maturity.⁸

2.2. Related literature and contributions

Our study contributes to three strands of literature. First, our paper contributes to the literature on the stock pledge. Stock pledge exists in many markets around the world, including some major developed markets.⁹ It has unique characteristics compared to traditional credit financing from the following aspects: First, traditional credit financing typically requires physical collaterals while stock pledge does not. Therefore, it provides a much easier way especially for small firms and intangible assets intensive firms to access to external debts; Second, it offers firms a lower cost of capital because the lenders can quickly sell the pledged shares in case that the margin call is triggered, which reduces the risk borne by the lenders. Third, it brings a significant risk of losing controlling rights to the controlling shareholders if a margin call is triggered by a stock price slump. The uniqueness of the stock pledge might cause critical economic consequences.

However, there is no systematic research on stock pledge yet, and most of the existing studies focus only on the effects of stock pledge on near-term firm value and shareholders' wealth. For example, [Dou et al. \(2019\)](#) find that stock pledge by firm insiders for personal loans in Taiwan increases the firm's stock crash risk and document a negative causal relationship between the stock pledge and outside shareholders' wealth. Similarly, [Anderson and Puleo \(2015\)](#) find that insiders' stock pledge significantly increases firm's risk in the U.S. On the other hand, [Li et al. \(2019\)](#) find that stock pledge by major shareholders in China has a positive impact on firm value, although this relationship is moderated by state ownership and high ownership concentration. These studies, however, have ignored the long-term impact of stock pledge on firms and economic growth. Our paper adds to the literature by providing the first rigorous empirical analysis to examine the effects of stock pledge on innovation, which is a critical driver of long-term growth and competitiveness of firms and the national economy. Our findings have significant policy implications concerning the regulations of the stock pledge not only to China, but also to many other countries with similar concentrated ownership structure as China.

Second, our paper contributes to the vast literature on motivating innovation. There is ample empirical evidence to show that manager's myopia towards short-term performance due to career concerns stifles innovations while alleviating manager's career concern spurs innovation, which is consistent with the predictions of [Manso \(2011\)](#) and [Ederer and Manso \(2013\)](#). For instance, [Stein \(1988\)](#) and [Shleifer and Summers \(1988\)](#) argue that hostile takeover pressure leads to fewer managerial incentives to invest in innovation because managers have to keep stocks from being undervalued. [Lerner et al. \(2011\)](#) find that firms going private through LBO tend to be more innovative because private equity relieves managers from short-term pressure from public shareholders. [Aghion et al. \(2013\)](#) find that higher institutional ownership improves innovation because institutional investors focus on long-term commitment so that they can reduce the short-term career risk of the manager. [Fang et al. \(2014\)](#) find that stock liquidity impedes innovation because high stock liquidity makes firms more prone to hostile takeovers.

However, the conclusions of above literature are based on a presumption that the safety of manager's career is determined by the board, and the manager's behavior is likely to be disciplined by the takeover market, which are typical characteristics of the diffused ownership structure. However, in markets with concentrated ownership, the tight control creates an entrenchment that may allow the manager (controlling shareholder) to go unchallenged internally by the board of directors or externally by takeover markets. From the perspective of agency problem, several studies show that concentrated ownership fosters innovation because it alleviates high agency and contracting costs associated with innovation ([Francis and Smith, 1995](#); [Hall and Lerner, 2010](#)). But they ignore the possibility that the controlling shareholders may also have risk of losing corporate control even though they are so entrenched, for example, in case of the forced share sale triggered by the margin call of pledged stocks. The prevalent stock pledge by controlling shareholders in China provides us a unique opportunity to tackle this issue.

Third, our paper also adds to the literature on finance and innovation. Much of the literature on finance and innovation documents that innovative firms are more likely to face financial constraints (e.g., [Hall, 2002](#); [Brown et al., 2009](#); [Acharya and Xu, 2017](#)). Following literature has shown that the development of financial market affects innovation ([Nanda and Nicholas, 2014](#); [Hsu et al., 2014](#)). Most studies achieve a consensus that the development of external equity financing markets fosters innovation ([Brown et al., 2009](#); [Hsu et al., 2014](#)), but the opinions towards the effect of the development of credit financing on innovation are mixed.¹⁰ However, the main focus of existing literature is on the effects of traditional equity and credit markets (i.e., bank loans and corporate debts) on firm's growth opportunities and innovation, but is silent on other non-traditional financing channels, such as stock pledge. Although stock pledge is a kind of credit financing in nature, its unique characteristics may bring different effects on firm growth and innovation compared to traditional credit financing. This study enriches this literature by filling this gap.

⁸ According to our conversations with practitioners, the pledger may borrow about 40%–50% of the market value of pledged shares, and the maintenance requirements are typically 150% of the pledged amount. Once the maintenance margin is triggered, the pledger is either required to provide more shares as collateral or top up the margin account with cash. If the pledger fails to do so and the stock value further decreases to 140% of the pledged amount, the shares will be forced sold. For example, a stock with 100 RMB market value can be pledged for about 50 RMB loan, and the maintenance requirement is $100 \times 50\% \times 150\% = 75$ RMB. As long as the stock price decreases by 25%, the margin call will be triggered. If the pledger fails to top up the margin account and the stock price continues to decrease to $100 \times 50\% \times 140\% = 70$ RMB, the share will be forced sold.

⁹ For example, stock pledge by corporate insiders is allowed in U.S., U.K., Japan, Australia, Singapore, Hong Kong, Taiwan, and etc.

¹⁰ Although some early studies point strongly against the role of bank loans and corporate debts in financing innovation ([Hall, 1994](#); [Opler and Titman, 1994](#)), subsequent works tend to be more nuanced. [Nanda and Nicholas \(2014\)](#) show that bank distress negatively affects firm-level innovation and conclude that bank loans are important to innovation while [Hsu et al. \(2014\)](#) argue that the development of credit markets impedes innovation using an international data. [Cornaggia et al. \(2015\)](#) further find that more bank loans reduce state-level innovation but spur innovation for firms with limited access to external credit markets.

3. Data and descriptive statistics

3.1. Sample construction

Our initial sample covers all A-share listed firms in China from 2003 to 2017. We obtain stock pledge data from the China Stock Market & Accounting Research (CSMAR) Database. CSMAR collects the number of shares that still under pledging for the top 10 largest shareholders each year from firms' annual reports. We further obtain controlling shareholders information, ownership of controlling and other large shareholders, firms' patents data, stock return, and financial data, SOE status, listed dates, industry classifications and geographic locations from CSMAR and supplement institutional ownership information from Wind Terminal. To make our analyses more specific, we exclude firms which have no controlling shareholders and firms whose controlling shareholders are not the largest shareholders.¹¹

Our final sample consists of 19,122 firm-year observations for 2424 non-financial firms from 2003 to 2013, including 1371 non-SOEs and 1053 SOEs.¹² We start our sample period from 2003 because stock pledge information is not available before it. Our sample ends in 2013 because we want to mitigate the truncation problem of patents data. The truncation problem refers to a mechanical decrease in the number of patent granted as one approaches the end of the sample period, because there is a significant lag between a patent's application year and its grant year. It is worth to note that our patents data ends in 2014, which is one year ahead of stock pledge data and other control variables since we want to examine the effects of stock pledge on firm's future innovation outputs.

Following innovation literature, we control for a set of firm characteristics that may affect a firm's innovation output in our analyses. The detailed descriptions and definitions of variables used in this study are provided in [Appendix A](#).

3.2. Measuring innovation

We use patent rather than R&D expenditures to measure innovation activities because of three reasons: First, following the argument from [He and Tian \(2013\)](#), R&D expenditures only capture the observed innovation inputs but ignore unobserved inputs; Second, Chinese firms only voluntarily disclose their R&D expenditures. Therefore, a missing R&D expenditure does not necessarily imply that the firm has no innovation activity. Last, the R&D expenditures disclosed by Chinese firms are lack of comparability because there is no standardized reporting requirement.

There are three types of patents granted in China: invention patents, utility model patents, and design patents. Invention patents are granted for a new revolutionary solution to a product, and utility model patents are granted for new and practical solutions related to the shape or structure of a product ([Tan et al., 2015](#)). The design patents protect the appearance of a product and have a much lower inventiveness level compared to invention patents and utility patents. Therefore, we construct our main innovation outputs measures using a firm's (including its subsidiaries) total number of invention patents and utility patents application in year t that are eventually granted ($PatGr_t$).

Although we limit our patent data sample period to 2014 to avoid serious truncation problem, [Fig. 1](#) shows a gradual decrease in the number of patent applications that are eventually granted as we approach the last two years in the sample period. It suggests that the truncation problem may still exist to a certain extent in the year 2013 and 2014, especially for invention patents. It is because that it could take up to 5 years for an invention patent to finally get approval, but we can only get the patent granted information till 2017 from CSMAR. Therefore, following [Hall et al. \(2001, 2005\)](#) and [Fang et al. \(2014\)](#), we correct for this truncation bias by first calculating the lag distribution for patents filed and granted between 2003 and 2012. The application-grant lag distributions W_s is calculated as the percentage of patents applied for in a given year that are granted in s years. Since invention patents usually take a much longer time to be granted than utility patents, we calculate their lag distributions separately. We define W_{si} as the lag distribution for invention patents and W_{su} as the lag distribution for utility patents. We then compute the truncation-adjusted patent counts for invention patents as $InvPatGr_{t,adj} = \frac{InvPatGr_t}{\sum_{s=0}^{2017-t} W_{si}}$ and for utility patents as $UtiPatGr_{t,adj} = \frac{UtiPatGr_t}{\sum_{s=0}^{2017-t} W_{su}}$, where $InvPatGr_t$ ($UtiPatGr_t$) is the raw number of the invention (utility) patent granted in year t in our dataset and $2013 \leq t \leq 2014$. The truncation adjusted total number of patents granted in year t equals the summation of $InvPatGr_{t,adj}$ and $UtiPatGr_{t,adj}$. Following existing innovation literature, we use the natural logarithm of 1 plus patent counts in our analysis to mitigate the concerns to the potential skewness of the measure.

A potential concern about this variable is that it measures only the quantity but not the quality of innovation. Innovation literature using U.S. data frequently uses the number of patents' future citations as a measure of innovation quality. However, the patent citation information is not available in China. Therefore, following [Tan et al. \(2015\)](#) and [Yuan and Wen \(2018\)](#), we use the number of invention patents application in year t and eventually granted ($InvPatGr_t$) as an alternative measure of innovation quality. Similarly, we use the natural logarithm of 1 plus invention patent counts in our analysis to address skewness issue.

3.3. Sample distribution and descriptive statistics

[Table 1](#) reports our sample distribution of stock pledges and innovations across the years. Panel A of [Table 1](#) presents the distribution of firms with stock pledge across time. We can observe that the numbers of firms with stock pledge by either controlling shareholders or

¹¹ In our sample, only 73 firms (307 firm-year) have no controlling shareholders (because the largest shareholders ownership is too small) and 14 firms (62 firm-year) have different largest shareholders and controlling shareholders.

¹² Since institutional investors may have very different attitudes towards innovation compared with individuals, we have also checked the types of controlling shareholders. We find that the firms in our final sample are either controlled by the governments or by the individuals. None of them is controlled by institutional investors.

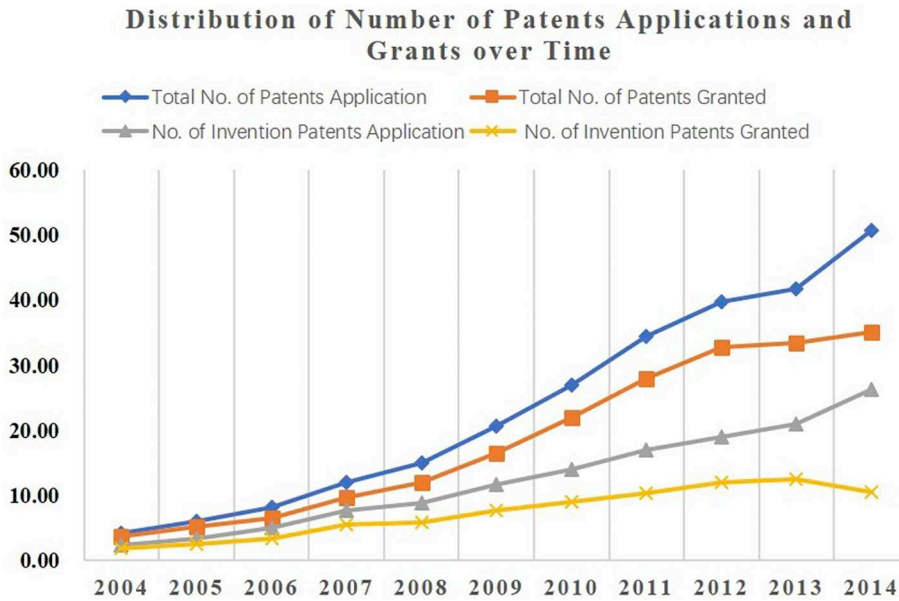


Fig. 1. Truncation problem of patent data.

This figure illustrates the potential truncation problem of our raw patent data used in the study.

non-controlling shareholders are increasing over time. The percentage of firms with stock pledge by controlling shareholders increases from 21.5% (266 out of 1238) in 2003 to 37.4% (907 out of 2424) in 2013, and the percentage of firms with stock pledge by non-controlling shareholders increases from 19.9% in 2003 to 31.6% in 2013. This result indicates that the stock pledge becomes more and more prevalent in China. Panel B of Table 1 shows the distribution of the percentage of shares held and pledged by block shareholders each year. The percentage of shares pledged over firms' total shares outstanding increases from 12.1% in 2003 to 18.5% in 2013. Moreover, the percentage of shares pledged over the total number of shares held by controlling (non-controlling) shareholders increases from 13.6% (7.1%) in 2003 to 23.2% (9.2%) in 2013. These findings show that shareholders pledge their shares more aggressively over time, and the controlling shareholders have more willingness to pledge their shares compared to non-controlling shareholders. We also observe that the average percentage of shares held by the controlling shareholders in our sample is more than 35%, while the average percentage of shares held by all non-controlling shareholders is only around 20%. Although the ownership of controlling shareholders seems to decrease over time, the data still suggests that the controlling shareholders in China tend to be very entrenched.

Panel C of Table 1 presents the average number of patents applications and grants per firm each year. During the sample period, the average numbers of both patents applications and grants per firm increased significantly, indicating that Chinese firms are transitioning fast from low-cost manufacturing enterprises to higher value innovation-led enterprises. Moreover, more originated and revolutionary innovation outputs are generated by Chinese firms, reflected by the significant increase of invention patents applications and grants.¹³

Table 2 presents the descriptive summary statistics for our sample. To address the concerns about the outliers, we winsorize all variables at the 1st and 99th percentiles.

Our sample firms on average generate 22.48 patents each year, and 9.58 of them are invention patents. 28% of the controlling shareholders in our sample have stock pledge experience and on average, they pledge 18% of shares they hold in exchange for loans. Compared to controlling shareholders, less percentage of non-controlling shareholders pledge their shares. Regarding other control variables, an average firm has been listed for eight years and has a leverage of 49%, ROA of 3%, book-to-market ratio of 1, and capital expenditure ratio of 6%.

4. Main empirical results

4.1. Baseline results

To investigate how stock pledge by controlling shareholder affects innovation, first, we estimate the following model:

$$PatGrt_{i,t+n}(InvPatGrt_{i,t+n}) = \alpha + \beta * IfTop1Pledge_{i,t} + \gamma * Controls_{i,t} + Firm_i + Year_t + \varepsilon_{i,t} \quad (1)$$

¹³ The strong economic growth of China during our sample period allowed the government to spend more on education. More educated workforces laid a solid foundation for innovation activities. Moreover, since China joins in WTO in 2001, Chinese firms began to realize the importance of intellectual properties to the global competition so that they started to invest more in R&D. Last but not least, the Chinese government significantly enforces the protection of intellectual property rights after joining the WTO.

Table 2

Descriptive statistics.

This table reports the summary statistics for variables constructed based on the sample of A-share listed firms in China from 2003 to 2014. All variables are winsorized at 1% and 99% levels to mitigate the impact of outliers. The statistics of *PatApp*, *PatGrt*, *InvPatApp* and *InvPatGrt* in this table are calculated based on the original numbers of total patent applications, total patent granted, invention patent applications and invention patent granted, respectively. Definitions of other variables are listed in [Appendix A](#).

	N	Mean	SD	25th Pctl	Median	75th Pctl
<i>PatApp</i>	19,122	27.37	177.36	0.00	2.00	13.00
<i>PatGrt</i>	19,122	22.48	140.89	0.00	1.00	10.00
<i>InvPatApp</i>	19,122	14.17	127.56	0.00	0.00	5.00
<i>InvPatGrt</i>	19,122	9.58	90.60	0.00	0.00	3.00
<i>IfTop1Pledge</i>	19,122	0.28	0.45	0.00	0.00	1.00
<i>IfNonTop1Pledge</i>	19,122	0.23	0.42	0.00	0.00	0.00
<i>Top1PledgePer</i>	19,101	0.18	0.33	0.00	0.00	0.22
<i>NonTop1PledgePer</i>	19,121	0.07	0.17	0.00	0.00	0.00
<i>MTop1Per_NoPat</i>	16,699	0.19	0.26	0.00	0.08	0.30
<i>Assets</i>	19,119	21.54	1.23	20.72	21.41	22.22
<i>R&D</i>	19,122	0.06	0.18	0.00	0.00	0.06
<i>SalesGrowth</i>	17,946	0.23	0.62	-0.02	0.14	0.31
<i>Leverage</i>	19,119	0.49	0.27	0.31	0.48	0.63
<i>Tangibility</i>	19,119	0.26	0.18	0.12	0.23	0.38
<i>Capex</i>	19,083	0.06	0.06	0.02	0.04	0.09
<i>CFO</i>	19,119	0.04	0.08	0.00	0.04	0.09
<i>ROA</i>	19,119	0.03	0.07	0.01	0.03	0.06
<i>B/M</i>	18,704	1.01	0.85	0.44	0.75	1.30
<i>Age</i>	19,122	8.04	5.37	3.00	8.00	12.00
<i>Ret</i>	18,870	0.28	0.93	-0.27	-0.01	0.52
<i>Vol</i>	18,984	0.03	0.05	0.02	0.03	0.04
<i>IO</i>	19,122	0.19	0.22	0.01	0.09	0.32
<i>SOE</i>	19,122	0.57	0.50	0.00	1.00	1.00
<i>Reform</i>	19,122	0.47	0.50	0.00	0.00	1.00

where i indicates firm, t indicates year, and n equals to 1 or 2. The dependent variables are the natural logarithm of 1 plus total number of patents applied and eventually granted in year $t + n$ and the natural logarithm of 1 plus number of invention patents applied and eventually granted in year $t + n$, respectively. Because the innovation process usually takes more than one year, we examine the effect of stock pledge by controlling shareholders on the innovation in subsequent years. *IfTop1Pledge_{it}* is a dummy variable which indicates the existence of stock pledge by controlling shareholder of firm i in year t . *Controls_{it}* is a set of control variables that may affect a firm's future innovation outputs. *Firm_{it}* and *Year_t* capture the firm and year fixed effects. The robust standard errors are clustered at firm-level to account for any correlations among firms.

We control for firm fixed effects in our baseline regressions in order to mitigate the concern about unobserved time-invariant omitted variables that affect both stock pledge and innovation. For example, over-confident controlling shareholders may be more likely to pledge shares as they are more confident about future firm performance and stock price. Besides, over-confident controlling shareholders may also actively engage in innovation projects. In this case, the controlling shareholders' personality is unobserved and time-invariant but correlated with both stock pledge and innovation, which could bias our coefficient estimation in the analyses.

Table 3 reports the impact of the existence of stock pledge on corporate innovation from estimating Eq. (1) by using OLS regressions. The dependent variable in column (1) is the total number of patent applications filed and eventually granted in the following year (*PatGrt_{t+1}*). The coefficient of *IfTop1Pledge* is negative and significant at the 5% level, suggesting that the existence of controlling shareholder's stock pledge decreases the level of innovation output in the following year. Compared with the firms without controlling shareholder's stock pledge, the firms with controlling shareholder's stock pledge are associated with a 4.91% decrease in the number of total patents granted in the following year. However, the coefficient of *IfNonTop1Pledge* is insignificant, suggesting that the existence of non-controlling shareholders' stock pledge has an insignificant impact on innovation output. This finding is consistent with [Chan et al. \(2018\)](#)'s argument that when losing corporate control is not a concern to blockholders, the stock pledge should exhibit no relationship with corporate decisions. Consistent with existing literature, we also find that larger and older firms, or firms with more growth opportunities and more tangible assets have more innovation outputs. Moreover, institutional ownership is significantly positively related to innovation which is consistent with the findings in [Aghion et al. \(2013\)](#) and [He and Tian \(2013\)](#).

In column (2), we replace the dependent variable with the total number of patent application filed and eventually granted in the year $t + 2$ (*PatGrt_{t+2}*). However, the coefficients of *IfTop1Pledge* turns to be insignificantly negative, suggesting that controlling shareholders are more likely to cut down short-term innovation rather than long-term innovation projects. It is reasonable because short-term innovation projects may have more impact on near-term firm performance and stock prices, and controlling shareholders have a stronger incentive to mitigate short-term stock price crash risk.

In column (3) and (4), we replace the dependent variable with the number of invention patents filed and eventually granted, *InvPatGrt_{t+n}*. Since there is no patent citation information available in China, we expect the number of invention patents granted to capture the quality of innovation to some extent. We find that the coefficients of *IfTop1Pledge* are both negative and significant in

Table 3

Existence of stock pledge by controlling shareholder and innovation.

This table reports pooled regressions of the innovation output variables on stock pledge indicators and other control variables. $PatGrt_{t+n}$ ($InvPatGrt_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year $t + n$. Dummy variable $IfTop1Pledge$ equals to 1, and zero otherwise, if the firm's controlling shareholder has outstanding shares under pledge at the end of year t . All the other control variables are defined in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)
IfTop1Pledge	−0.0491** (−2.17)	−0.0260 (−1.15)	−0.0425** (−2.34)	−0.0332* (−1.83)
IfNonTop1Pledge	−0.0047 (−0.17)	0.0024 (0.09)	−0.0069 (−0.30)	−0.0120 (−0.54)
Assets	0.2788*** (14.98)	0.2475*** (13.49)	0.2233*** (14.88)	0.2064*** (13.81)
R&D	0.7385*** (5.61)	0.5820*** (5.76)	0.7234*** (6.12)	0.5846*** (5.73)
SalesGrowth	−0.0307*** (−2.78)	−0.0240** (−2.22)	−0.0291*** (−3.59)	−0.0286*** (−3.73)
Leverage	−0.1016* (−1.87)	−0.0923* (−1.66)	−0.0091 (−0.23)	−0.0017 (−0.04)
Tangibility	0.3196*** (4.64)	0.4162*** (5.91)	0.2795*** (5.29)	0.3220*** (5.93)
Capex	0.0110 (0.08)	−0.0491 (−0.34)	0.1295 (1.15)	0.0238 (0.21)
CFO	−0.0823 (−0.94)	−0.0113 (−0.13)	0.0130 (0.19)	−0.0039 (−0.06)
ROA	0.0440 (0.38)	−0.0033 (−0.03)	0.0069 (0.08)	−0.0202 (−0.23)
B/M	−0.0637*** (−4.29)	−0.0664*** (−4.55)	−0.0569*** (−4.69)	−0.0654*** (−5.56)
Age	0.2866*** (7.52)	0.2619*** (7.00)	0.2274*** (7.03)	0.1929*** (6.08)
Ret	−0.0265** (−2.35)	−0.0157 (−1.48)	−0.0230** (−2.50)	−0.0247*** (−2.79)
Vol	−0.0443 (−0.21)	0.1106 (0.65)	−0.0282 (−0.18)	0.2033* (1.81)
IO	0.1416*** (3.03)	0.0826* (1.78)	0.1137*** (2.97)	0.0792** (2.08)
SOE	−0.0226 (−0.64)	−0.0264 (−0.73)	0.0302 (1.13)	−0.0047 (−0.16)
Reform	−0.0049 (−0.13)	0.0009 (0.02)	−0.0369 (−1.18)	−0.0212 (−0.64)
Constant	−5.5702*** (−14.30)	−4.8048*** (−12.55)	−4.7090*** (−15.06)	−4.2072*** (−13.54)
Year Fixed Effects	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes
Observations	17,535	17,480	17,535	17,480
R-squared	0.812	0.826	0.789	0.807

column (3) and (4). However, the coefficient of $IfTop1Pledge$ in column (3) is more significant with a larger magnitude than the coefficient in column (4), indicating that short-term innovation quality is hurt more by the stock pledge. Moreover, the coefficients of $IfNonTop1Pledge$ remain insignificant in both columns. Therefore, we can conclude that the existence of controlling shareholder's stock pledge is significantly negatively related to the firms' future innovation outputs and quality, but the non-controlling shareholders' pledging behavior does not affect innovation.

Investigating the effects of the percentage of shares pledged by the controlling shareholder on innovation may make more economic senses. The more percentage of shares the controlling shareholder has pledged, the more likely he is to lose corporate control if the stock price crashes. Therefore, we use the percentage of shares pledged by controlling shareholder as a continuous variable and estimate the following model:

$$PatGrt_{i,t+n} (InvPatGrt_{i,t+n}) = \alpha + \beta * Top1PledgePer_{i,t} + \gamma * Controls_{i,t} + Firm_i + Year_t + \varepsilon_{i,t} \quad (2)$$

where $Top1PledgePer_{i,t}$ is the ratio of shares pledged over the total shares held by the controlling shareholder of firm i in year t . Other variables share the same definitions with those in Eq. (1).

Table 4 reports the impact of the percentage of shares pledged by controlling shareholder on corporate innovation from estimating

Eq. (2) by using OLS regressions. The coefficients of *Top1PledgePer* are negative and significant for all columns, suggesting that a higher percentage of shares pledged by the controlling shareholder is associated with a lower level of innovation outputs and quality in the future. For example, in column (1), the coefficient of *Top1PledgePer* is -0.0813 indicating that 1% increase in shares pledged by the controlling shareholder is associated with 0.0813% decrease in the number of patents granted in the following year. However, we find that both the magnitude and significance of the coefficient of *Top1PledgePer* in column (2) are lower than those of the coefficient in column (1), suggesting that the percentage of pledged shares by controlling shareholder affects short-term innovation outputs more. Similarly, we find that the coefficient of *Top1PledgePer* in column (3) has a larger magnitude than that in column (4), although they are both significant at 1% level. Overall, according to the results in Table 4, we can conclude that the percentage of shares pledged by the controlling shareholder is significantly negatively related to the firm's future innovation outputs and quality.

4.2. Robustness checks

We also perform a series of additional tests to ensure that the significant negative relationship between stock pledge by controlling shareholder and innovation is robust to alternative sample selections, model specifications, and variable definitions. First, we would like to investigate whether stock pledge affects 3-year-ahead innovation outputs. Panel A of Table 5 shows *Top1PledgePer* becomes insignificantly related to 3-year-ahead total patent outputs, while it is still significantly related to 3-year-ahead invention patent outputs at 1% level. It is not weird to see that the percentage of stock pledge has a diminishing effect on the total number of patents granted over time but has a relatively persistent effect on the number of invention patents granted. As we can see from Table 1 Panel C, on average the number of utility patents granted each year is two times more than the number of invention patents granted, because it is much faster and less risky for firms to generate utility patents than invention patents. Therefore, the effect of stock pledge on invention patent outputs lasts longer than that on utility patent outputs.

Second, we note that there is a portion of firms in our sample never generate any patent. The controlling shareholders of those non-innovative firms may pledge their shares more aggressively, so that the negative relation between the stock pledge and innovation we observed in our baseline regressions may be driven by the subsample of non-innovative firms.¹⁴ To address this concern, we exclude all firms that never generate any patent or invention patent and repeat the analyses from the model (2). The results are shown in Panel B of Table 5, and we find that the negative relation between the stock pledge and innovation is still robust.

Moreover, there is a particular group of stocks existing in China market that are called ST (Special Treatment) stocks, because they report losses for consecutive two fiscal years. Firms with poorer performance usually are less innovative. In addition, those poorly performed firms usually are more financially constrained, so the controlling shareholders of ST firms may have more incentive to pledge their shares. Therefore, the negative relationship we observed between the stock pledge and innovation may be driven by the subsample of ST firms. To address this concern, we exclude all ST firms from our sample and repeat the analyses from the model (2). The results are shown in Panel C of Table 5, and we find that the relationship between the stock pledge and innovation is still robust.

We have previously mentioned the truncation problem of our sample, and we have attempted to correct this problem by adopting a commonly used method in the existing literature. However, Dass et al. (2017) show that the current methods used to correct truncation problem of U.S. patent data work poorly. It is also possible that the method we adopted may not solve the problem for Chinese patent data, leading to biased results. To mitigate this concern, we use an even shorter sample period which is end at 2011 and repeat previous tests. Panel D of Table 5 presents the regression results in the shorter sample, and we find that the relationship between the stock pledge and innovation is not affected.

Listed firms are located in different provinces in China. It is plausible that the unique characteristics of each province may affect both the stock pledge and innovation, rendering our findings spurious. Therefore, we add province fixed effects in our regression model, and the results in Panel E of Table 5 show that the relationship between the stock pledge and innovation is not affected.

Although we use the number of granted patents as the measure of innovation in this study, we admit that the number of patents application also matters. More patents application may reflect more R&D inputs from a firm. Therefore, we use the number of total patents application (*PatApp*) as an alternative measure of innovation quantity, and the number of invention patents application (*InvPatApp*) as an alternative measure of innovation quality and repeat the tests. The results in Panel F of Table 5 show that the coefficients of *Top1PledgePer* are always negative and significant, which is consistent with what we observed for the relationship between the stock pledge and the number of granted patents.

4.3. Propensity score matching approach

Empirical research using multiple regressions always suffers from functional form misspecification bias (Rosenbaum and Rubin, 1983; Shipman et al., 2016) and produces biased estimation. Propensity score matching is a powerful method to mitigate functional form misspecification bias. Following Rosenbaum and Rubin (1983), we perform a 1-to-1 nearest-neighbor matching of firms whose controlling shareholders have stocks under pledge (treatment group) with firms whose controlling shareholders do not pledge stocks

¹⁴ More specifically, we find that in our sample, 588 firms never generate any patent and 760 firms never generate any invention patent. We thank the referee for pointing out this possibility.

Table 4

Percentage of shares pledged by controlling shareholders and innovation.

This table reports pooled regressions of the innovation output variables on the percentage of shares pledged by controlling shareholder and other control variables. $PatGr_{t+n}$ ($InvPatGr_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year $t + n$. $Top1PledgePer$ is the percentage of shares pledged by controlling shareholders at the end of year t . All the other control variables are defined in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

	PatGr _{t+1}	PatGr _{t+2}	InvPatGr _{t+1}	InvPatGr _{t+2}
	(1)	(2)	(3)	(4)
Top1PledgePer	−0.0813*** (−3.00)	−0.0473* (−1.73)	−0.0803*** (−3.76)	−0.0574*** (−2.73)
NonTop1PledgePer	−0.0423 (−0.95)	−0.0161 (−0.37)	−0.0081 (−0.23)	−0.0362 (−1.06)
Asset	0.2805*** (15.16)	0.2478*** (13.57)	0.2239*** (14.97)	0.2065*** (13.84)
R&D	0.7652*** (5.64)	0.5956*** (5.69)	0.7490*** (6.15)	0.5978*** (5.64)
SalesGrowth	−0.0315*** (−2.86)	−0.0244** (−2.26)	−0.0299*** (−3.68)	−0.0291*** (−3.80)
Leverage	−0.0986* (−1.83)	−0.0902 (−1.63)	−0.0066 (−0.16)	0.0002 (0.00)
Tangibility	0.3185*** (4.62)	0.4149*** (5.89)	0.2770*** (5.24)	0.3222*** (5.93)
Capex	−0.0003 (−0.00)	−0.0499 (−0.34)	0.1252 (1.11)	0.0186 (0.16)
CFO	−0.0814 (−0.93)	−0.0115 (−0.13)	0.0172 (0.25)	−0.0035 (−0.05)
ROA	0.0416 (0.36)	−0.0011 (−0.01)	0.0039 (0.04)	−0.0208 (−0.24)
B/M	−0.0660*** (−4.45)	−0.0663*** (−4.55)	−0.0581*** (−4.80)	−0.0653*** (−5.56)
Age	0.2849*** (7.49)	0.2622*** (7.01)	0.2257*** (6.99)	0.1911*** (6.03)
Ret	−0.0270** (−2.39)	−0.0159 (−1.50)	−0.0230** (−2.51)	−0.0250*** (−2.82)
Vol	−0.0522 (−0.25)	0.1051 (0.62)	−0.0343 (−0.22)	0.1985* (1.79)
IO	0.1369*** (2.93)	0.0793* (1.71)	0.1116*** (2.92)	0.0769** (2.02)
SOE	−0.0310 (−0.86)	−0.0318 (−0.88)	0.0218 (0.81)	−0.0107 (−0.37)
Reform	−0.0035 (−0.09)	0.0020 (0.05)	−0.0350 (−1.12)	−0.0205 (−0.61)
Constant	−5.5973*** (−14.47)	−4.8074*** (−12.63)	−4.7130*** (−15.15)	−4.2010*** (−13.56)
Year Fixed Effects	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes
Observations	17,514	17,459	17,514	17,459
R-squared	0.812	0.826	0.789	0.807

(control group) each year based on their propensity scores. More specifically, we construct the matched sample of firms first by running a probit regression of a dummy variable that equals to one and zero otherwise if a firm's controlling shareholder has shares under pledge at the year-end, on all control variables in our baseline regressions as well as the industry and year dummies in the full sample. We control for year and industry dummies to capture any time-invariant or industry-specific differences. We then calculate the predicted propensity score for each firm using the estimated coefficients from the probit estimation. Last, we use the calculated propensity scores to perform a nearest-neighbor matching by selecting one control firm for one treatment firm with the closest propensity score. Finally, we obtain 5195 treatment observations and their corresponding control observations.

Panel A of Table 6 reports the univariate difference of innovation outputs between the treatment group and the control group. On average, the total number of patent granted in the following year of the treatment group (1.181) is significantly lower than that of the control group (1.370). We observe similar significant average patent outputs differences between the treatment group and control group for $PatGr_{t+2}$, $InvPatGr_{t+1}$, and $InvPatGr_{t+2}$. These results are consistent with our earlier findings that firms with the existence of controlling shareholder's stock pledge are less innovative than those without it. To further control the difference of characteristics between treatment group and control group, we perform multivariable regressions in the matched sample by regressing each innovation output variable on the treatment dummy and the same control variables in our baseline regressions with the year and firm fixed effects controlled. As shown in Panel B of Table 6, the coefficients of the treatment dummy are negative and significant across

Table 5

Robustness checks on alternative samples, model specifications and variable definitions.

This table reports pooled regressions of the innovation output variables on the percentage of shares pledged by controlling shareholder and other control variables by using different samples, model specifications and variable definitions. $PatGr_{t+n}$ ($InvPatGr_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year $t + n$. $Top1PledgePer$ is the percentage of shares pledged by controlling shareholder at the end of year t . Panel A lags the covariates by 3 years. Panel B excludes firms that never generate any patent from the sample. Panel C excludes ST stocks from the sample. Panel D uses the sample from 2003 to 2011. Panel E controls the province fixed effects. Panel F uses the number of patent applications filed (including both granted and non-granted patents) in a year as the alternative measure of innovation output. All the other control variables are the same as those in Table 4 and their definitions are provided in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Panel A: lag the covariates by 3 Years				
	PatGrt _{t+3}	InvPatGrt _{t+3}		
	(1)	(2)		
Top1PledgePer	−0.0411 (−1.48)	−0.0620*** (−2.81)		
Panel B: excluding firms never generate any patent				
	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)
Top1PledgePer	−0.0956** (−2.37)	−0.0529 (−1.30)	−0.1027*** (−2.90)	−0.0691** (−1.98)
Panel C: excluding ST stocks				
	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)
Top1PledgePer	−0.0977*** (−3.15)	−0.0651** (−2.09)	−0.0929*** (−3.78)	−0.0717*** (−2.99)
Panel D: sample ends at 2011				
	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)
Top1PledgePer	−0.0605** (−2.10)	−0.0270 (−0.92)	−0.0719*** (−3.20)	−0.0446** (−1.97)
Panel E: control for province fixed effects				
	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)
Top1PledgePer	−0.0813*** (−3.00)	−0.0473* (−1.73)	−0.0803*** (−3.76)	−0.0574*** (−2.73)
Panel F: use total number of patents application in a year as the measure of innovation				
	PatApp _{t+1}	PatApp _{t+2}	InvPatApp _{t+1}	InvPatApp _{t+2}
	(1)	(2)	(3)	(4)
Top1PledgePer	−0.0936*** (−3.24)	−0.0559* (−1.89)	−0.0847*** (−3.44)	−0.0719*** (−2.94)
Control Variables	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes

Table 6

Propensity score matching.

This table reports the effects of the existence of controlling shareholder's stock pledge on innovation output by using propensity score matching method. The control group is selected based on a nearest-neighbor matching of propensity scores from a probit regression. Panel A reports the univariate difference of innovation output between the treatment group and control group. Panel B reports the multivariable regressions in the matched sample by regressing each innovation output variable on treatment dummy, which equals one, and zero otherwise, if the firm's controlling shareholder pledge the shares at the end of year t . $PatGrt_{t+n}$ ($InvPatGrt_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year $t + n$. All the other control variables are the same as those in Table 4 and their definitions are provided in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Panel A: difference in future innovation between treatment and control groups.					
	N	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
Treatment	5195	1.181	1.273	0.731	0.808
Control	5195	1.370	1.472	0.886	0.972
Difference		-0.189*** (-6.05)	-0.199*** (-6.12)	-0.155*** (-6.42)	-0.164*** (-6.47)

Panel B: regression analysis.				
	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
Treatment	-0.0449* (-1.68)	-0.0446* (-1.67)	-0.0494** (-2.26)	-0.0492** (-2.25)
Control Variables	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes
Observations	10,390	10,390	10,390	10,390
R-squared	0.830	0.844	0.812	0.830

the models, which provides supporting evidence that stock pledge by controlling shareholder stifles the firm's innovation.¹⁵

4.4. Instrumental variable approach

Although we have addressed certain concerns with endogeneity issues about the negative relationship between the stock pledge and innovation through controlling firm fixed effects and performing propensity score matching, it is still possible that our findings are caused by some other omitted variables and even by reverse causality. It is plausible that the controlling shareholders of low innovation potential firms pledge their shares more aggressively. To further mitigate the endogeneity concerns, we construct an instrumental variable for the percentage of shares pledged by controlling shareholder and use the 2SLS approach to correct potential bias. The ideal instrumental variable should be relevant to the variation in the percentage of shares pledged by controlling shareholder but be exogenous to the firm's innovation productivity.

We construct the instrumental variable *MTop1Per_NoPat*, as the “average percentage of shares pledged by the controlling shareholders of non-innovative firms (firms never generate any patent) that are located in the same province and belong to the same industry”.¹⁶ We require the firms to be located in the same province because similar regional availability of financing may induce the controlling shareholders in the same province to adopt similar stock pledge strategy. For example, Firth et al. (2009) find that firms located in more developed provinces in China are more likely to receive bank loans. Furthermore, local governments in China always provide direct government subsidies or tax rebates to locally listed firms (Cull et al., 2017), but the amount of the subsidies may be different across provinces. We also require the firms to belong to the same industry because Rajan and Zingales (1998) find that different industries have different demand for external financing, which may affect the controlling shareholders' stock pledge incentives. We understand that firms in the same industry may have similar innovation incentive and the knowledge spillover effects within geographic industry cluster may also affect firms' innovation outputs, so we only use non-innovative firms when constructing the instrumental variable to mitigate the incentive similarity and knowledge spillover effects (Chang et al., 2015).

Results obtained using instrumented 2SLS approach are presented in Table 7. Column (1) of Table 7 presents the first-stage regression result with the percentage of shares pledged by controlling shareholders (*Top1PledgePer*) as the dependent variable to

¹⁵ We also perform a 1-to-3 nearest-neighbor matching and report the results in Table OA1 in the Online Appendix. The results are consistent.

¹⁶ We follow the CSRC Guidelines for Classification of listed Companies in China (CSRC Classification) to define the industry for Chinese listed firms. The construction of CSRC classification learns from SIC and NAICS, but uses different classification code. For example, a code range “2000–3999” refers to Manufacturing division in SIC, and the corresponding code in CSRC Classification is a letter ‘C’; a 2-digit number ‘36’ refers to a business that deals in “Electronic and Other Equipment” in SIC, and the corresponding code in CSRC Classification is ‘C85’. Since most literature on U.S. market uses 2-digit number of SIC code to define the same industry, we use “a letter plus 2-digit number” of CSRC classification to define the same industry in this study.

check the relevance of the instrument. All the other control variables are the same as those in the baseline regression model (2). Year and firm fixed effects are controlled, and standard errors are clustered at the firm level. The coefficient of the instrumental variable, *MTop1Per_NoPat* is positive and significant at 1% level with a large t-value, suggesting that it has a strong correlation with *Top1PledgePer*. We further conducted the weak identification test (Kleibergen-Paap Wald F Statistics) and find that the F-Statistic is 477.43 and significant at 1%. This result suggests that our instrumental variable is not a weak instrument. Column (2) to (5) presents the regression results of the second stage. Similar to the results we observed in the OLS regressions, the coefficients of *Top1PledgePer* are always negative and significant.

However, we take care in interpreting the 2SLS results as strong evidence for a causal effect of stock pledge by controlling shareholders on innovation because our instrument may not be perfect. For example, zero-patent firms can still be innovative if they choose not to patent their innovation outputs. Ideally, we should directly check whether our instrument satisfies exogenous criteria using econometrical methods. However, to our best knowledge, currently available econometrical methods only allow us to check the exogeneity of the instrumental variables for overidentified equations (i.e. more instrumental variables than the endogenous variables). Since we only have one instrumental variable, our equation is exactly identified so that we are not able to conduct a direct exogeneity test.

In summary, while endogeneity is one of the most challenging issues in empirical research that no empirical tests can absolutely rule out, we conduct several tests to alleviate the endogeneity concerns, and find that our main results hold. The balance of evidence should suggest a causal relation between stock pledge by controlling shareholders and innovation.

4.5. Mechanism of impediment effects of stock pledge on innovation

Although we find that stock pledge by controlling shareholder has a negative causal effect on innovation, the mechanism through which stock pledge impedes innovation is still ambiguous. We propose two different scenarios that may exacerbate controlling shareholder's concerns about losing corporate control and hypothesize that if the impediment effect of stock pledge on innovation is caused by the controlling shareholder's fear of losing corporate control, it should be more pronounced in these scenarios.

The first scenario is that the controlling shareholder's position is potentially challenged by the second-largest shareholder, i.e., the ownership of controlling shareholder is close to the ownership of the second-largest shareholder. If a controlling shareholder does not hold a wide lead in ownership over the second-largest shareholder, he may feel a stronger sense of crisis about being replaced. Therefore, he may tend to make more conservative corporate decisions that substantially reduce the firm's risk exposure to keep his pledged shares safe.

The second scenario is that the stock price crash risk of a firm increases significantly around the stock pledge. A stock price crash during the stock-pledge period would seriously threaten the safety of pledged shares. If the margin call is triggered by a stock price slump and the controlling shareholder is not able to top up his margin account, the pledged shares will be forced sold and the controlling shareholder may lose his controlling position. Kothari et al. (2009) and Hutton et al. (2009) suggest that an increase in a firm's stock price crash risk might indicate that the manager of the firm was hiding bad news. The controlling shareholder has an incentive to hide negative news before the stock pledge because a higher market price of the stock will help him to get more loans. However, Jin and Myers (2006) and Kim et al. (2011) find that if the accumulation of bad news passes a critical threshold level, it will be revealed to the market at one time and lead to a stock price crash. Therefore, to keep his pledged shares and controlling position safe, the controlling shareholder has a strong incentive to avoid generating bad news as much as possible during the stock-pledging period to avoid further bad news stockpiling. However, He and Wong (2004) and Jia (2018) argue that innovation activities have a high failure rate and tend to generate more bad news, which may accelerate the stock price crash. Consequently, the controlling shareholder who has experienced a large increase in stock price crash risk tends to be more conservative in investing in innovative projects, if he has shares under pledge.

We create a controlling power competition index (CPC-index) which is the ratio between the ownership of the largest and second-largest shareholders for each firm-year. A lower CPC-index indicates that the controlling shareholder faces more potential controlling rights challenges. We then sort our sample by the CPC-index and partition the sample by the median. We define the subsample of below-median CPC index as higher controlling power competition subsample and the subsample of above-median CPC index as lower controlling power competition subsample. We estimate the model (2) in each subsample and the regression results are reported in Table 8. Column (1) to (4) of Table 8 present the results in higher controlling power competition subsample and column (5) to (8) show the results in lower controlling power competition subsample. We find that the coefficients of *Top1PledgePer* are negative and significant across models in higher controlling power competition subsample. On the other hand, the coefficients of *Top1PledgePer* are always negative but insignificant across models in lower controlling power competition subsample. We also observe that the magnitudes of coefficients are larger in higher controlling power competition subsample than the magnitudes of coefficients in corresponding models in lower controlling power competition subsample. Moreover, we directly compare the coefficients of *Top1PledgePer* in corresponding models from each subsample and find that their differences are statistically significant. These results suggest that the impediment effect of stock pledge on innovation is more pronounced if the controlling shareholder feels a stronger sense of crisis about being replaced.

Furthermore, following Chen et al. (2001) and Xu et al. (2014), we use the negative coefficients of skewness of firm-specific weekly returns (*ncskew*) to measure a firm's stock price crash risk. To construct this measure, we first estimate firm-specific weekly return $R_{i,t}$ as the natural log of one plus the residual return from the following model:

$$r_{i,j} = \alpha_i + \beta_1 * R_{m,j-2} + \beta_2 * R_{m,j-1} + \beta_3 * R_{m,j} + \beta_4 * R_{m,j+1} + \beta_5 * R_{m,j+2} + \varepsilon_{i,j} \quad (3)$$

Table 7

Two-stage least-squares regressions.

This table reports the 2SLS regressions of the innovation output variables on the percentage of shares pledged by controlling shareholder using an instrumental variable approach. $PatGrt_{t+n}$ ($InvPatGrt_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year $t + n$. $MTop1Per_NoPat$ is the instrumental variable which defined as the average percentage of shares pledged by controlling shareholders of non-innovative firms (firms never generate any patent) that are located in the same province and belong to the same industry. The table reports the results for both the first-stage regression, which generates the fitted value of $Top1PledgePer$ and the second-stage regressions using the fitted value of $Top1PledgePer$. All the other control variables are defined in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

	First-stage regression		Second-stage regressions		
	Top1PledgePer	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)	(5)
Top1PledgePer		−0.1917*** (−2.89)	−0.1133* (−1.76)	−0.1824*** (−3.38)	−0.1213** (−2.33)
MTop1Per_NoPat	0.5036*** (38.44)				
NonTop1PledgePer	0.2012*** (11.32)	−0.0163 (−0.34)	0.0130 (0.27)	0.0199 (0.53)	−0.0140 (−0.38)
Assets	0.0297*** (5.46)	0.2214*** (11.97)	0.1932*** (10.57)	0.1795*** (12.16)	0.1598*** (10.97)
R&D	−0.0439** (−2.34)	0.6699*** (4.88)	0.5102*** (4.92)	0.6694*** (5.38)	0.4976*** (4.86)
SalesGrowth	−0.0181*** (−4.96)	−0.0291*** (−2.91)	−0.0241** (−2.37)	−0.0276*** (−3.81)	−0.0252*** (−3.60)
Leverage	0.0114 (0.60)	−0.1285** (−2.40)	−0.1038* (−1.88)	−0.0247 (−0.64)	−0.0275 (−0.71)
Tangibility	−0.0393* (−1.86)	0.3256*** (4.92)	0.4368*** (6.37)	0.2586*** (5.28)	0.3280*** (6.37)
Capex	−0.0120 (−0.30)	0.0117 (0.08)	−0.0247 (−0.17)	0.1166 (1.07)	0.0571 (0.51)
CFO	0.0232 (0.80)	−0.1171 (−1.37)	−0.0350 (−0.39)	0.0069 (0.11)	−0.0359 (−0.54)
ROA	−0.0251 (−0.59)	0.0443 (0.40)	0.0574 (0.51)	0.0036 (0.04)	0.0440 (0.53)
B/M	−0.0035 (−0.90)	−0.0483*** (−3.23)	−0.0529*** (−3.51)	−0.0476*** (−3.97)	−0.0536*** (−4.59)
Age	0.0889*** (8.51)	0.3466*** (8.44)	0.3326*** (8.31)	0.2766*** (8.09)	0.2720*** (8.16)
Ret	0.0095*** (3.05)	−0.0165 (−1.48)	−0.0076 (−0.71)	−0.0133 (−1.49)	−0.0165* (−1.89)
Vol	−0.1178** (−1.98)	−0.1075 (−0.55)	0.0262 (0.15)	−0.0921 (−0.64)	0.1386 (1.20)
IO	−0.0129 (−1.04)	0.1005** (2.08)	0.0541 (1.12)	0.0850** (2.20)	0.0488 (1.25)
SOE	−0.1506*** (−10.92)	−0.0089 (−0.24)	−0.0109 (−0.29)	0.0269 (0.96)	−0.0119 (−0.40)
Reform	0.0320** (2.42)	−0.0030 (−0.08)	0.0220 (0.56)	−0.0408 (−1.36)	0.0022 (0.06)
F-test of Instruments	477.43 (0.00)	N.A.	N.A.	N.A.	N.A.
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes
Observations	15,055	15,055	15,020	15,055	15,020
R-squared	0.706	0.819	0.832	0.805	0.818

where $r_{i,j}$ is the return of stock i in week j and $R_{m,j}$ is the value-weighted A-share market return in week j . Then the firm-specific weekly return $R_{i,j} = \ln(1 + \varepsilon_{i,j})$, and $\varepsilon_{i,j}$ is the residual of Eq. (3).

$Ncskew_{i,t}$ is then calculated by taking the negative of the third moment of each stock i 's firm-specific weekly return for each year t and dividing it by the cubed standard deviation of firm-specific weekly returns. Thus, for any stock i over year t , we get Eq (4):

$$ncskew_{i,t} = \frac{-n(n-1)^{3/2} \sum R_{i,t}^3}{(n-1)(n-2)(\sum R_{i,t}^2)^{3/2}} \quad (4)$$

where n is the number of observations of firm-specific weekly returns during the year t . An increase in $ncskew$ corresponding to a stock being more “crash risk prone”, that is, having a more left-skewed distribution. We further calculate the change of $ncskew$,

Table 8

Impediment effects of stock pledge on innovation and controlling power competition.

This table reports pooled regressions of the innovation output variables on the percentage of shares pledged by controlling shareholder and other control variables in subsamples separated by firms' controlling power competition index. The controlling power competition index (CPC Index) is calculated as the ratio of ownership of controlling shareholder to the ownership of second largest shareholder in a firm. Higher (Lower) level of potential controlling power competition subsample includes the firms whose CPC Index is below (above) the median value. $PatGrt_{t+n}$ ($InvPatGrt_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year $t + n$. $Top1PledgePer$ is the percentage of shares pledged by controlling shareholder at the end of year t . All the other control variables are defined in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

	Higher Controlling Power Competition Subsample				Lower Controlling Power Competition Subsample			
	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Top1PledgePer	−0.1112*** (−3.14)	−0.0679* (−1.87)	−0.0878*** (−3.07)	−0.0742*** (−2.66)	−0.0234 (−0.49)	−0.0291 (−0.60)	−0.0486 (−1.33)	−0.0208 (−0.58)
NonTop1PledgePer	−0.1009* (−1.90)	−0.0662 (−1.27)	−0.0738* (−1.83)	−0.0949** (−2.36)	−0.0466 (−0.45)	−0.0142 (−0.13)	0.1016 (1.17)	0.1030 (1.18)
Assets	0.2898*** (10.44)	0.2506*** (9.30)	0.2330*** (10.18)	0.2142*** (9.39)	0.2877*** (9.24)	0.2784*** (8.93)	0.2268*** (9.14)	0.2190*** (8.76)
R&D	0.4359*** (3.66)	0.3269*** (3.76)	0.4388*** (4.24)	0.3035*** (3.66)	1.3859*** (9.42)	1.0090*** (6.68)	1.3699*** (9.49)	1.1197*** (7.37)
SalesGrowth	−0.0420*** (−2.95)	−0.0265* (−1.82)	−0.0270*** (−2.75)	−0.0305*** (−3.23)	−0.0053 (−0.33)	−0.0058 (−0.37)	−0.0176 (−1.35)	−0.0082 (−0.69)
Leverage	0.0547 (0.69)	0.0531 (0.65)	0.0424 (0.75)	0.0390 (0.71)	−0.2932*** (−3.26)	−0.2230** (−2.31)	−0.0870 (−1.29)	−0.0052 (−0.07)
Tangibility	0.4582*** (4.81)	0.5034*** (5.09)	0.3985*** (5.50)	0.3912*** (5.03)	0.2290** (2.16)	0.4262*** (3.82)	0.1500* (1.82)	0.2707*** (3.20)
Capex	0.0541 (0.28)	0.2288 (1.19)	0.1579 (1.03)	0.2645* (1.74)	0.1230 (0.57)	−0.2438 (−1.11)	0.2352 (1.45)	−0.1842 (−1.09)
CFO	−0.0351 (−0.28)	0.0460 (0.36)	0.0392 (0.41)	0.0293 (0.30)	−0.1243 (−0.96)	−0.0228 (−0.18)	−0.0237 (−0.25)	−0.0378 (−0.38)
ROA	0.2881** (1.98)	0.0410 (0.28)	0.1550 (1.34)	−0.0943 (−0.85)	−0.1099 (−0.54)	0.1077 (0.53)	−0.0577 (−0.40)	0.1858 (1.26)
B/M	−0.0728*** (−3.38)	−0.0592*** (−2.88)	−0.0530*** (−3.06)	−0.0490*** (−2.96)	−0.0435** (−2.04)	−0.0600*** (−2.78)	−0.0358** (−2.14)	−0.0636*** (−3.79)
Age	0.2178*** (4.08)	0.2601*** (5.00)	0.1928*** (4.24)	0.1959*** (4.40)	0.4033*** (6.58)	0.2972*** (4.82)	0.3093*** (6.12)	0.2199*** (4.33)
Ret	−0.0275* (−1.80)	−0.0164 (−1.15)	−0.0344*** (−2.64)	−0.0221* (−1.83)	−0.0215 (−1.32)	−0.0125 (−0.75)	−0.0080 (−0.62)	−0.0282** (−2.11)
Vol	−0.4464*** (−3.15)	0.1435 (0.88)	−0.2653** (−2.25)	0.2519* (1.79)	0.2975 (1.30)	0.0945 (0.29)	0.2870** (2.18)	0.2988 (1.57)
IO	0.0372 (0.52)	−0.0598 (−0.84)	0.0890 (1.46)	−0.0254 (−0.42)	0.2072*** (3.14)	0.1919*** (2.86)	0.1138** (2.14)	0.1559*** (2.93)
SOE	−0.0352 (−0.77)	−0.0400 (−0.85)	0.0054 (0.14)	−0.0297 (−0.73)	0.0705 (0.98)	0.0521 (0.72)	0.1317*** (2.77)	0.0891* (1.78)
Reform	0.0360 (0.73)	−0.0204 (−0.40)	−0.0071 (−0.18)	−0.0346 (−0.84)	−0.0472 (−0.77)	0.0177 (0.27)	0.0265 (−0.54)	0.0207 (0.39)
Constant	−5.6497*** (−9.71)	−4.8385*** (−8.55)	−4.7919*** (−10.03)	−4.3012*** (−8.99)	−6.1026*** (−9.40)	−5.6695*** (−8.73)	−5.0768*** (−9.85)	−4.7020*** (−9.04)
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8755	8732	8755	8732	8759	8727	8759	8727
R-squared	0.849	0.863	0.833	0.850	0.820	0.828	0.796	0.809

Comparison of Coefficients of $Top1PledgePer$ for Model (1) and (5): $\chi^2 = 3.23$, $p - value = 0.072$

Comparison of Coefficients of $Top1PledgePer$ for Model (2) and (6): $\chi^2 = 2.22$, $p - value = 0.137$

Comparison of Coefficients of $Top1PledgePer$ for Model (3) and (7): $\chi^2 = 2.72$, $p - value = 0.091$

Comparison of Coefficients of $Top1PledgePer$ for Model (4) and (8): $\chi^2 = 3.15$, $p - value = 0.075$

denoted as $\Delta ncskew_{i,t}$ in year t for stock i as the difference between $ncskew$ in year t ($ncskew_t$) and that in year $t-1$ ($ncskew_{t-1}$).

We then sort our sample by $\Delta ncskew$ and partition it into quartiles. We find that the average $\Delta ncskew$ from the top to the bottom quartiles are 1.29, 0.26, −0.28, and −1.18, respectively. Therefore, we define the top quartile as Large Increase in Stock Price Crash Risk subsample and the bottom quartile as Large Decrease in Stock Price Crash Risk subsample. We do not use the middle two quartiles because a small change in stock price crash risk around stock pledge may not significantly affect the controlling shareholders' fear of losing corporate control.

We estimate the model (2) in each subsample, and the regression results are reported in Table 9. Column (1) to (4) of Table 9 present the regression results in Large Increase in Stock Price Crash Risk subsample while column (5) to (8) show the results in Large

Decrease in Stock Price Crash Risk subsample. We find that the coefficients of *Top1PledgePer* are more negative and significant in Large Increase in Stock Price Crash Risk subsample than corresponding coefficients in Large Decrease in Stock Price Crash Risk subsample, suggesting that the impediment effect of stock pledge on innovation is stronger when the controlling shareholder is under greater stock price crash pressure.

Taken together, the findings in Tables 8 and 9 provide suggestive evidence that reveals the possible mechanism through which stock pledge impedes innovation. This impediment effect is likely due to the controlling shareholder's fear of losing corporate control.

4.6. Effects of stock pledge on financially constrained firms

Although we have found that in general, stock pledge by controlling shareholders impedes corporate innovation, it is still possible that stock pledge may foster innovations for financially constrained firms because it provides the firms an easier, faster, and less costly external financing channel. We next directly examine the effect of stock pledge on innovation for financially constrained firms. Following Hadlock and Pierce (2010), we construct a financial constraints index for each firm-year. We then partition our sample into quartiles based on the index and define the firms in the quartile with the highest index as higher financial constraint subsample, and remaining firms as lower financial constraint subsample.¹⁷ If financial constraints relaxation hypothesis is valid, we expect to observe a less negative or even positive relationship between the stock pledge and innovation in the higher financial constraint subsample. We estimate the model (2) in each subsample and the regression results are reported in Table 10. Column (1) to (4) of Table 10 present the results in higher financial constraint subsample and column (5) to (8) show the results in the lower financial constraint subsample. We find that the coefficients of *Top1PledgePer* are negative and significant in both subsamples, except for column (2) and column (6), suggesting that stock pledge by controlling shareholder impedes firms' innovation regardless of their level of financial constraints. Moreover, we directly compare the coefficients of *Top1PledgePer* in corresponding models from each subsample and find that the coefficients in higher financial constraint subsample are significantly more negative than corresponding coefficients in lower financial constraint subsample. These results contradict the implications of financial constraints relaxation hypothesis.¹⁸

However, it is possible that a large portion of the controlling shareholders in our sample pledge shares for personal loans rather than for corporate loans (Singh, 2018). If the loans obtained through stock pledge are not invested in the firm, the firm's financial constraints problem will not be relieved. Unfortunately, we are not able to directly investigate this issue because the information of the ultimate use of the loan proceeds is not publicly available. But we can conduct an indirect investigation within SOE subsample based on a reasonable presumption that all controlling shareholders of SOEs may only pledge shares for corporate loans.

We partition our sample of SOEs into quartiles based on their financial constraint indexes. We define SOEs in the quartile with the highest index as financially constrained SOEs and the remaining as non-financially constrained SOEs. We estimate the model (2) in each subsample and the regression results are presented in Table 11. Column (1) to (4) of Table 11 show the results in financially constrained SOEs subsample and column (5) to (8) show the results in non-financially constrained SOEs subsample. We find that the stock pledge has similar impediment effects on innovation for both financially-constrained and non-financially-constrained SOEs. The coefficients of *Top1PledgePer* are always negative across models in both subsamples, and the comparison of magnitudes of coefficients between corresponding models in each subsample are generally statistically insignificant. These results indicate that even though the shares are pledged for corporate loans, it still does not enhance the firm's innovation.

Taken all the above findings together, we can conclude that although stock pledge is a possible channel to relieve firms' financial constraints, it does not encourage the firm to invest more in innovation.

5. Conclusion

In this paper, we have examined the effects of stock pledge by controlling shareholder on corporate innovation. We find that both the existence of stock pledge by controlling shareholder and the percentage of shares pledged by controlling shareholder are significantly negatively related to firm's future innovation outputs and quality. The negative relationship is robust in different samples, model specifications and variable definitions. To address potential endogeneity concerns and to establish causality, we control for firm fixed effects in the regression analyses, perform propensity score matching, and adopt the instrumental variable approach. The identification tests suggest that stock pledge by controlling shareholder impedes innovation.

We further provide evidence to show that the impediment effect of stock pledge by controlling shareholder on innovation is more pronounced when the controlling shareholder faces more severe inside controlling power challenge and when the firm is under more stock price crash pressure. These findings suggest that the stock pledge impedes innovation is possibly due to controlling shareholder's fear of losing corporate control in case of innovation failure. Moreover, we provide evidence to show that although stock pledge by controlling shareholder is a possible channel to relieve a firm's financial constraint problem, it does not encourage more investments in innovation.

With the rapid increase in stock pledge in China markets, it should be a concern to government regulators, investors and firms'

¹⁷ We also attempt to partition our sample into tertiles and by the median, the results are consistent. Furthermore, we construct a KZ index following Kaplan and Zingales (1997) and Cao et al. (2018) as an alternative measure of financial constraints and the results are similar.

¹⁸ Since most financially constrained firms in China are non-SOE firms, we also separate our sample into SOEs and non-SOEs subsamples, and find that the stock pledge has similar impediment effects on innovation for SOEs and non-SOEs. The results are presented in Table OA2 in the Online Appendix.

Table 9

Impediment effects of stock pledge on innovation and change in stock price crash risk.

This table reports pooled regressions of the innovation output variables on the percentage of shares pledged by controlling shareholder and other control variables in subsamples separated by the change in stock price crash risk of a firm. The stock price crash risk, *ncskew* is constructed following Chen et al. (2001). The change of stock price crash risk in year *t*, $\Delta ncskew_t$ is calculated as $ncskew_t - ncskew_{t-1}$. The Large Increase in Stock Price Crash Risk subsample includes the firms whose $\Delta ncskew_t$ is in the top quartile, while the Large Decrease in Stock Price Crash Risk subsample includes firms whose $\Delta ncskew_t$ is in the bottom quartile. $PatGrt_{t+n}$ ($InvPatGrt_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year *t* + *n*. *Top1PledgePer* is the percentage of shares pledged by controlling shareholder at the end of year *t*. All the other control variables are defined in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

	Large increase in stock price crash risk subsample				Large decrease in stock price crash risk subsample			
	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Top1PledgePer	-0.1964** (-2.39)	-0.0922 (-1.08)	-0.1778*** (-2.95)	-0.0465 (-0.73)	-0.0248 (-0.37)	0.0408 (0.57)	-0.0237 (-0.45)	-0.0165 (-0.31)
NonTop1PledgePer	-0.1072 (-0.85)	-0.0457 (-0.36)	-0.1038 (-1.11)	-0.1107 (-1.10)	-0.0693 (-0.59)	-0.0539 (-0.48)	-0.0582 (-0.67)	-0.0495 (-0.63)
Assets	0.2204*** (4.05)	0.1978*** (3.84)	0.1677*** (4.12)	0.1574*** (3.86)	0.2838*** (6.29)	0.2086*** (4.24)	0.2093*** (5.85)	0.1560*** (3.84)
R&D	0.4291 (1.56)	0.4840** (2.09)	0.5201** (2.28)	0.4731* (1.89)	1.7744*** (6.64)	1.1767*** (3.75)	1.6916*** (6.64)	1.3088*** (4.68)
SalesGrowth	-0.0366 (-1.13)	-0.0270 (-0.87)	-0.0386* (-1.82)	-0.0303 (-1.29)	-0.0745** (-2.35)	-0.0603* (-1.85)	-0.0555** (-2.28)	-0.0315 (-1.37)
Leverage	0.0356 (0.23)	-0.1133 (-0.70)	0.0309 (0.28)	0.0384 (0.33)	0.0281 (0.20)	0.1250 (0.95)	0.0490 (0.51)	0.1250 (1.34)
Tangibility	0.4314* (1.92)	0.5576** (2.52)	0.3424** (2.28)	0.3529** (2.17)	0.3326* (1.95)	0.2528 (1.43)	0.1404 (1.11)	0.2360* (1.70)
Capex	-0.2665 (-0.64)	0.1413 (0.33)	0.0041 (0.01)	0.0332 (0.10)	0.2445 (0.60)	0.1725 (0.42)	0.1592 (0.52)	0.1604 (0.48)
CFO	-0.2087 (-0.88)	0.0176 (0.07)	0.0300 (0.16)	0.0598 (0.31)	-0.2079 (-0.91)	0.0812 (0.36)	-0.0558 (-0.33)	-0.0181 (-0.11)
ROA	0.6104* (1.74)	0.2704 (0.75)	0.1543 (0.61)	0.1546 (0.57)	0.0941 (0.31)	0.0583 (0.20)	0.0004 (0.00)	-0.0273 (-0.13)
B/M	-0.0212 (-0.50)	-0.0319 (-0.71)	-0.0261 (-0.77)	-0.0227 (-0.65)	-0.0840** (-2.32)	-0.0901** (-2.44)	-0.0790*** (-2.84)	-0.0767*** (-2.60)
Age	0.2508** (2.55)	0.2463** (2.43)	0.2561*** (3.22)	0.1189 (1.44)	0.5060*** (3.93)	0.3650*** (2.75)	0.3595*** (3.33)	0.2733** (2.39)
Ret	-0.0398 (-1.16)	-0.0414 (-1.24)	-0.0312 (-1.16)	-0.0170 (-0.62)	-0.0304 (-1.21)	-0.0360 (-1.38)	-0.0375* (-1.88)	-0.0389* (-1.85)
Vol	1.0499 (0.67)	-1.1229 (-0.62)	-0.3712 (-0.33)	-0.6931 (-0.45)	1.1707* (1.93)	0.7777 (1.19)	1.0383** (2.20)	0.5767 (0.97)
IO	-0.1211 (-0.86)	-0.0838 (-0.60)	-0.0554 (-0.50)	-0.0225 (-0.20)	0.0788 (0.59)	0.0827 (0.63)	0.0798 (0.75)	0.1012 (0.93)
SOE	0.0365 (0.37)	0.0278 (0.29)	0.0450 (0.63)	0.0068 (0.09)	0.1610* (1.77)	0.0891 (0.93)	0.1337** (2.31)	0.1069 (1.55)
Reform	0.1144 (1.11)	0.0602 (0.53)	0.0263 (0.30)	0.0689 (0.68)	0.1864* (1.89)	0.1499 (1.52)	0.1201 (1.41)	0.0299 (0.32)
Constant	-4.3567*** (-3.82)	-3.6925*** (-3.45)	-3.5639*** (-4.23)	-3.0632*** (-3.67)	-6.3046*** (-6.74)	-4.4238*** (-4.33)	-4.7418*** (-6.40)	-3.4472*** (-4.08)
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4287	4274	4287	4274	4297	4289	4297	4289
R-squared	0.878	0.885	0.869	0.875	0.878	0.882	0.865	0.871

Comparison of Coefficients of *Top1Pledgeper* for Model (1) and (5): $\chi^2 = 4.91$, *p* - value = 0.027

Comparison of Coefficients of *Top1Pledgeper* for Model (2) and (6): $\chi^2 = 2.71$, *p* - value = 0.095

Comparison of Coefficients of *Top1Pledgeper* for Model (3) and (7): $\chi^2 = 7.10$, *p* - value = 0.008

Comparison of Coefficients of *Top1Pledgeper* for Model (4) and (8): $\chi^2 = 0.35$, *p* - value = 0.552

Table 10

Stock pledge, innovation and financial constraints.

This table reports pooled regressions of the innovation output variables on the percentage of shares pledged by controlling shareholder and other control variables in subsamples separated by the firms' financial constraint index, which is calculated following Hadlock and Pierce (2010). The higher financial constraint subsample includes firms whose financial constraint index is in the top quartile, while the lower financial constraint subsample includes remaining firms. $PatGrt_{t+n}$ ($InvPatGrt_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year $t + n$. $Top1PledgePer$ is the percentage of shares pledged by controlling shareholder at the end of year t . All the other control variables are defined in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

	Higher financial constraint subsample				Lower financial constraint subsample			
	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}	PatGrt _{t+1}	PatGrt _{t+2}	InvPatGrt _{t+1}	InvPatGrt _{t+2}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Top1PledgePer	-0.2135** (-2.39)	-0.1286 (-1.51)	-0.2504*** (-3.43)	-0.1446** (-2.08)	-0.0527* (-1.87)	-0.0281 (-0.99)	-0.0495** (-2.29)	-0.0393* (-1.82)
NonTop1PledgePer	0.0175 (0.12)	0.1238 (0.91)	-0.0981 (-0.78)	-0.2004* (-1.77)	-0.0326 (-0.71)	-0.0227 (-0.49)	0.0112 (0.32)	-0.0155 (-0.45)
Assets	0.4968*** (7.50)	0.3658*** (5.55)	0.4731*** (8.22)	0.3410*** (6.02)	0.2493*** (10.74)	0.2233*** (9.69)	0.1864*** (10.60)	0.1690*** (9.47)
R&D	0.9507*** (4.66)	0.7678*** (3.71)	1.0883*** (4.85)	0.9705*** (4.34)	0.5928*** (4.63)	0.4529*** (4.63)	0.5540*** (5.20)	0.4116*** (4.52)
SalesGrowth	-0.0090 (-0.34)	0.0004 (0.02)	-0.0139 (-0.64)	-0.0021 (-0.12)	-0.0197* (-1.70)	-0.0184 (-1.54)	-0.0150* (-1.82)	-0.0204*** (-2.58)
Leverage	-0.2639 (-1.08)	-0.0827 (-0.34)	-0.2472 (-1.15)	-0.0735 (-0.37)	-0.0953* (-1.66)	-0.0922 (-1.54)	-0.0186 (-0.46)	-0.0128 (-0.31)
Tangibility	0.4323** (2.27)	0.4095** (2.09)	0.3107** (2.09)	0.3540** (2.33)	0.2093*** (2.84)	0.3791*** (5.02)	0.2358*** (4.19)	0.2970*** (5.05)
Capex	-0.2006 (-0.55)	-0.5239 (-1.39)	0.0249 (0.08)	-0.3105 (-1.02)	0.0798 (0.52)	0.1699 (1.11)	0.2494** (2.10)	0.2349** (1.98)
CFO	-0.1440 (-0.64)	-0.0113 (-0.05)	-0.1587 (-0.86)	-0.2067 (-1.13)	-0.1291 (-1.38)	-0.0185 (-0.19)	-0.0279 (-0.40)	0.0018 (0.02)
ROA	0.4096 (0.73)	0.3015 (0.57)	0.4928 (1.00)	0.4648 (1.06)	0.1664 (1.44)	0.0892 (0.74)	0.0836 (1.00)	0.0368 (0.42)
B/M	-0.0644** (-2.45)	-0.0631** (-2.44)	-0.0717*** (-3.18)	-0.0848*** (-4.05)	-0.0624*** (-2.90)	-0.0727*** (-3.39)	-0.0416*** (-2.70)	-0.0478*** (-2.97)
Age	0.1492 (1.55)	0.2188** (2.27)	0.0285 (0.34)	0.1132 (1.40)	0.2990*** (6.93)	0.2868*** (6.68)	0.2770*** (7.88)	0.2424*** (6.82)
Ret	-0.0071 (-0.28)	-0.0070 (-0.26)	-0.0021 (-0.10)	-0.0165 (-0.77)	-0.0103 (-0.81)	0.0002 (0.02)	-0.0038 (-0.37)	-0.0055 (-0.56)
Vol	-1.2687* (-1.92)	-1.5854*** (-2.74)	-1.2750** (-2.40)	-1.2544*** (-2.65)	-0.1920 (-0.97)	0.0088 (0.05)	-0.1687 (-1.23)	0.1075 (0.88)
IO	0.1737** (2.10)	0.1535* (1.86)	0.0982 (1.46)	0.0715 (1.09)	0.0903 (1.55)	0.0282 (0.49)	0.0789* (1.68)	0.0418 (0.87)
SOE	-0.0541 (-0.28)	-0.1569 (-0.81)	-0.1033 (-0.71)	-0.1470 (-1.11)	-0.0263 (-0.71)	-0.0113 (-0.30)	0.0256 (0.96)	0.0106 (0.37)
Reform	0.0518 (0.45)	0.0888 (0.71)	0.0662 (0.64)	0.0238 (0.22)	-0.0045 (-0.11)	-0.0139 (-0.34)	-0.0334 (-1.05)	-0.0118 (-0.35)
Constant	-10.3529*** (-6.98)	-7.3556*** (-5.01)	-9.9614*** (-7.78)	-7.0161*** (-5.61)	-4.9236*** (-10.36)	-4.3463*** (-9.21)	-3.9816*** (-11.06)	-3.5091*** (-9.60)
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4379	4357	4379	4357	13,135	13,102	13,135	13,102
R-squared	0.885	0.895	0.879	0.894	0.785	0.801	0.748	0.767

Comparison of Coefficients of *Top1Pledgeper* for Model (1) and (5): $\chi^2 = 3.68$, p -value = 0.055

Comparison of Coefficients of *Top1Pledgeper* for Model (2) and (6): $\chi^2 = 1.56$, p -value = 0.211

Comparison of Coefficients of *Top1Pledgeper* for Model (3) and (7): $\chi^2 = 8.68$, p -value = 0.003

Comparison of Coefficients of *Top1Pledgeper* for Model (4) and (8): $\chi^2 = 2.60$, p -value = 0.107

Table 11

Stock pledge for corporate loans, financial constraints, and innovation.

This table reports pooled regressions of the innovation output variables on the percentage of shares pledged by controlling shareholder and other control variables in subsamples separated by the SOEs' financial constraint index, which is calculated following Hadlock and Pierce (2010). The financially-constrained SOEs subsample includes SOEs whose financial constraint index is in the top quartile, while the non-financially-constrained SOEs subsample includes the remaining SOEs. $PatGr_{t+n}$ ($InvPatGr_{t+n}$) is the natural logarithm of one plus the total number of patent (invention patent) applications filed and eventually granted in year $t + n$. $Top1PledgePer$ is the percentage of shares pledged by controlling shareholder at the end of year t . All the other control variables are defined in Appendix A. Each regression includes year and firm fixed effects. The robust standard errors are clustered by firms. The t-statistics are presented in the parenthesis and superscripts ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

	Financially-constrained SOEs				Non-financially-constrained SOEs			
	PatGr _{t+1}	PatGr _{t+2}	InvPatGr _{t+1}	InvPatGr _{t+2}	PatGr _{t+1}	PatGr _{t+2}	InvPatGr _{t+1}	InvPatGr _{t+2}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Top1PledgePer	-0.2923** (-1.98)	-0.1589 (-1.12)	-0.2347* (-1.85)	-0.1360 (-1.09)	-0.0484 (-0.92)	-0.0090 (-0.17)	-0.0615 (-1.45)	-0.0047 (-0.11)
NonTop1PledgePer	-0.0752 (-0.47)	0.0091 (0.05)	-0.0176 (-0.10)	-0.1852 (-1.18)	-0.0693 (-1.22)	-0.0912 (-1.58)	-0.0010 (-0.02)	-0.0488 (-1.15)
Asset	0.5104*** (5.87)	0.3105*** (3.62)	0.4803*** (6.35)	0.3109*** (4.29)	0.2651*** (8.16)	0.2236*** (6.69)	0.2039*** (7.79)	0.1843*** (6.76)
R&D	0.8364*** (3.27)	0.5594** (1.99)	0.8361*** (2.64)	0.7695** (2.40)	1.4689*** (8.88)	1.1334*** (6.97)	1.2951*** (9.23)	1.1231*** (7.30)
SalesGrowth	-0.0125 (-0.34)	0.0037 (0.12)	-0.0119 (-0.40)	0.0039 (0.17)	-0.0014 (-0.08)	-0.0250 (-1.41)	-0.0041 (-0.30)	-0.0248* (-1.86)
Leverage	-0.1849 (-0.54)	0.0338 (0.10)	-0.1514 (-0.49)	0.0447 (0.16)	-0.0615 (-0.73)	-0.0810 (-0.90)	-0.0789 (-1.22)	-0.0635 (-0.93)
Tangibility	0.3302 (1.41)	0.2871 (1.19)	0.3019* (1.66)	0.2997 (1.63)	0.3051*** (3.12)	0.4092*** (4.00)	0.3181*** (4.23)	0.3343*** (4.15)
Capex	-0.2922 (-0.59)	-0.1270 (-0.26)	-0.0118 (-0.03)	-0.1095 (-0.28)	0.0663 (0.33)	0.1132 (0.57)	0.1129 (0.75)	0.0999 (0.64)
CFO	-0.4477 (-1.47)	-0.6043** (-1.98)	-0.5312** (-2.15)	-0.4813* (-1.92)	-0.1870 (-1.46)	0.0107 (0.08)	-0.1326 (-1.35)	-0.0800 (-0.78)
ROA	0.4102 (0.53)	0.2820 (0.42)	0.7512 (1.09)	0.2999 (0.52)	0.3689** (2.10)	0.2573 (1.41)	0.1178 (0.94)	0.1324 (0.98)
B/M	-0.0965*** (-2.87)	-0.0776** (-2.32)	-0.0922*** (-3.22)	-0.1031*** (-3.72)	-0.0716*** (-2.82)	-0.0849*** (-3.31)	-0.0461** (-2.40)	-0.0579*** (-2.95)
Age	0.1168 (0.93)	0.2708** (2.20)	0.0234 (0.21)	0.1077 (1.02)	0.3969*** (6.13)	0.2938*** (4.58)	0.2717*** (5.00)	0.2203*** (4.12)
Ret	-0.0513 (-1.53)	0.0012 (0.03)	-0.0254 (-0.93)	-0.0144 (-0.52)	-0.0221 (-1.33)	-0.0089 (-0.54)	-0.0079 (-0.61)	-0.0251* (-1.88)
Vol	1.3290 (1.61)	0.3641 (0.16)	0.5886 (0.86)	0.5591 (0.28)	0.3085 (1.32)	-0.4627 (-1.43)	0.0015 (0.01)	0.0509 (0.22)
IO	0.1742* (1.68)	0.0289 (0.27)	0.1075 (1.29)	-0.0059 (-0.07)	-0.0715 (-0.96)	-0.0302 (-0.40)	-0.0029 (-0.05)	-0.0085 (-0.14)
Reform	0.0508 (0.34)	0.1142 (0.72)	0.0642 (0.47)	0.0201 (0.14)	0.0750 (1.37)	0.0282 (0.50)	-0.0106 (-0.23)	-0.0065 (-0.13)
Constant	-11.0636*** (-5.54)	-6.5101*** (-3.34)	-10.5933*** (-6.11)	-6.6777*** (-4.06)	-5.7706*** (-8.57)	-4.6177*** (-6.68)	-4.5098*** (-8.36)	-3.9456*** (-7.00)
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2584	2566	2584	2566	7748	7728	7748	7728
R-squared	0.884	0.897	0.879	0.895	0.763	0.774	0.725	0.741

Comparison of Coefficients of *Top1Pledgeper* for Model (1) and (5): $\chi^2 = 2.94$, $p - value = 0.086$

Comparison of Coefficients of *Top1Pledgeper* for Model (2) and (6): $\chi^2 = 1.22$, $p - value = 0.270$

Comparison of Coefficients of *Top1Pledgeper* for Model (3) and (7): $\chi^2 = 2.06$, $p - value = 0.151$

Comparison of Coefficients of *Top1Pledgeper* for Model (4) and (8): $\chi^2 = 1.21$, $p - value = 0.271$

management that stock pledge by controlling shareholder may lead to underinvestment in innovation. This could ultimately affect the health and long-term growth of firms and the national economy.

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Appendix A. Variable definitions

Variable	Definition
Measures of Innovation	
$PatGrt_{t+n}$	Natural logarithm of one plus firm's total number of invention and utility model patent applications filed (and eventually granted) in year $t + n$, where $n = 1, 2$ and 3 respectively.
$InvPatGrt_{t+n}$	Natural logarithm of one plus firm's total number of invention patent applications filed (and eventually granted) in year $t + n$, where $n = 1, 2$ and 3 respectively.
$PatApp_{t+n}$	Natural logarithm of one plus firm's total number of invention and utility model patent applications filed in year $t + n$, where $n = 1$ and 2 respectively.
$InvPatApp_{t+n}$	Natural logarithm of one plus firm's total number of invention patent applications filed in year $t + n$, where $n = 1$ and 2 respectively.
Measures of Stock Pledge	
$IfTop1Pledge_t$	A dummy variable which equals to one if the firm's controlling shareholder pledges the shares at the end of year t , otherwise it equals to zero.
$IfNonTop1Pledge_t$	A dummy variable which equals to one if the firm's non-controlling shareholders pledge their shares at the end of year t , otherwise it equals to zero.
$Top1PledgePer_t$	The number of shares pledged divided by the number of shares held by the controlling shareholder for a particular firm at the end of year t .
$NonTop1PledgePer_t$	The total number of shares pledged divided by the total number of shares held by the non-controlling shareholders for a particular firm at the end of year t .
$MTop1Per_NoPat_t$	The average percentage of shares pledged by the controlling shareholders of non-innovative firms (firms that never generate any patent) that are located in the same province and belong to the same industry at the end of year t .
Control Variables and Other Variables	
$Assets_t$	Natural logarithm of firm's total assets at the end of year t .
$R\&D_t$	R&D Expense/Total assets at the end of year t . Set to 0 if missing.
$SalesGrowth_t$	Annual sales growth rate which is calculated as $(Sales_t - Sales_{t-1})/Sales_{t-1}$.
$Leverage_t$	Total debt/Total assets at the end of year t .
$Tangibility_t$	PP&E/Total assets at the end of year t .
$Capex_t$	Capital expenditure/Total assets at the end of year t .
CFO_t	Operating cash flow/Total assets at the end of year t .
ROA_t	Net income of year t /Total assets at the end of year $t-1$.
B/M_t	Book value of the firm's equity/market value of the equity at the end of year t .
Age_t	The number of years the firm is listed.
Ret_t	The stock's buy and hold return in year t .
Vol_t	The standard deviation of the stock's daily returns in year t .
IO_t	Total institutional ownership which is calculated as the number of shares held by institutional shareholders divided by number of shares outstanding for a particular firm at the end of year t .
SOE_t	A dummy variable which equals to one if the firm's controlling shareholder is the government at the end of year t , otherwise it equals to zero.
$Reform_t$	A dummy variable which equals to one if a firm has completed the Split Share Structure Reform at the end of year t , otherwise it equals to zero.
$\Delta nskew_t$	The change of stock price crash risk of a firm in year t , which equals to $nskew_t - nskew_{t-1}$. $Nskew_t$ measures the negative coefficients of skewness of firm-specific weekly returns in year t , which is constructed following Chen et al. (2001).
$SA\ Index_t$	SA index is constructed following Hadlock and Pierce (2010) as $-0.737 * \logasset + 0.043 * (\logasset)^2 - 0.04 * age$ at the end of year t . It measures the level of a firm's financial constraint.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcorpfin.2019.101534>.

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